

The empirical recurrence for OEIS A317740, recovered from its tail

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Abstract

OEIS A317740 records “Empirical recurrence of order 97” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1702$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 12$, so the recurrence is proved for every $n \geq 109$. Its last coefficient is $c_{97} = -432 \neq 0$, so no further tail satisfies anything shorter; any order-97 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A317740 is “Number of $n \times 7$ 0..1 arrays with every element unequal to 2 or 3 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

$$0, 22, 6, 35, 74, 36, 274, 280, \dots$$

The entry states:

Empirical recurrence of order 97 (see link above)

As of the “Last modified” line on the live entry (Aug 05 2018 16:00:16, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1702$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1702$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 97: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 12$ is the first whose minimal order is exactly the stated 97.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3404$ exact terms from $n = 12$ on, it returns order 97 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{97} c_i a(n-i),$$

c_1	0	c_{26}	2436151	c_{51}	−982700325	c_{76}	48672959
c_2	6	c_{27}	−1256556	c_{52}	−134296466	c_{77}	−64434621
c_3	21	c_{28}	−7242654	c_{53}	821389450	c_{78}	24691365
c_4	−9	c_{29}	−4250779	c_{54}	1123712600	c_{79}	−21857085
c_5	−114	c_{30}	11234756	c_{55}	−384082812	c_{80}	18882385
c_6	−159	c_{31}	22262252	c_{56}	−559813510	c_{81}	−1714141
c_7	223	c_{32}	7549298	c_{57}	−906444618	c_{82}	−4136968
c_8	941	c_{33}	−36576632	c_{58}	846739744	c_{83}	1617125
c_9	289	c_{34}	−47923126	c_{59}	−124256516	c_{84}	−1257940
c_{10}	−2442	c_{35}	−9115416	c_{60}	638375406	c_{85}	1274082
c_{11}	−4559	c_{36}	86937446	c_{61}	−764445538	c_{86}	−191610
c_{12}	2666	c_{37}	82114510	c_{62}	587856017	c_{87}	−317224
c_{13}	15300	c_{38}	−25542651	c_{63}	−519495556	c_{88}	−66788
c_{14}	16252	c_{39}	−198878910	c_{64}	170606109	c_{89}	84764
c_{15}	−16864	c_{40}	−155188349	c_{65}	−211441861	c_{90}	96830
c_{16}	−62010	c_{41}	150482095	c_{66}	126813022	c_{91}	−2806
c_{17}	−61346	c_{42}	388584967	c_{67}	383124681	c_{92}	−21716
c_{18}	22615	c_{43}	285913511	c_{68}	−457340680	c_{93}	−6756
c_{19}	186556	c_{44}	−378271618	c_{69}	331027472	c_{94}	4488
c_{20}	272208	c_{45}	−590528267	c_{70}	−435253602	c_{95}	792
c_{21}	117825	c_{46}	−403451039	c_{71}	486996464	c_{96}	−1584
c_{22}	−531487	c_{47}	616554762	c_{72}	−307294248	c_{97}	−432
c_{23}	−995720	c_{48}	766657287	c_{73}	128309560		
c_{24}	−388684	c_{49}	393983599	c_{74}	−90588255		
c_{25}	1879191	c_{50}	−765223722	c_{75}	49337130		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 109$, and it is the recurrence OEIS A317740 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 12$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{97} - c_1x^{97-1} - \dots - c_{97}$. Its constant term is $-c_{97} = 432$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 97 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 97, so $\deg q = 97$, and it is monic. It holds from some index t on. From $\max(t, 12)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 97, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 30 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 97, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -432 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-97 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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